

Week 11 Deliverable

Integrated Finite-Element Analysis of the Chassis, Dump Bed, Hinge, and Actuator Mounts

Conceptual Design of a Lunar Regolith Transport Rover

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SOLIDWORKS Simulation linear-static screening of the final dump-bed mechanism and supporting chassis structure under nominal payload, doubled payload, raised-bed retained-load, abnormal jammed-load, and asymmetric terrain-support conditions. The work is a conceptual structural verification exercise, not flight qualification or final joint certification.

1 Week 11 Purpose

The Week 11 objective was to determine whether the final chassis support frame, upper deck, dump-bed shell, rear hinges, and dual-actuator mounting regions appear structurally adequate under the lunar payload and operating conditions established in earlier deliverables. The analysis used the actual integrated CAD geometry and the numerical results produced during the guided SOLIDWORKS Simulation workflow. Five static load cases were completed:

LC1: horizontal bed with the full 340 kg payload under lunar gravity;

LC2: horizontal bed with twice the payload pressure;

LC3: raised bed with 25 % retained payload;

LC4: raised bed with the full payload as an abnormal jammed-load condition; and

LC5: horizontal bed with asymmetric terrain-support boundary conditions.

Each study reports the model configuration, material, loads, fixtures, connectors, mesh, maximum von Mises stress, maximum resultant displacement, factor of safety (FOS), and reaction-force balance. The horizontal nominal case was also solved at coarse, medium, and fine mesh levels. Connector-force results were extracted for the raised cases and the final asymmetric-terrain case.

The report uses the same presentation style as the Week 8 preliminary FEA report [3]. The conclusion is intentionally limited to the completed load cases. In particular, these analyses do not replace a future full-rover chassis study in which the complete 990 kg loaded-rover design load is transferred through all suspension attachment regions.

2 Relationship to Prior Deliverables

Weeks 3 through 6 established the transport mission, rover envelope, mobility architecture, and rocker-bogie geometry. Week 7 performed a hand calculation of the dump-bed floor and recommended a 5 mm 6061-T6 aluminum sheet with transverse supports at no more than approximately 0.30 m spacing [2]. Week 8 screened selected rocker-bogie components by FEA [3]. Week 9 established the mission route and energy cycle. Week 10 completed the chassis, upper support frame, final dump-bed shell, rear hinges, actuator brackets, dual linear actuators, motion study, and interference checks [4].

Week 11 extends that work from isolated suspension components to the integrated dump-bed mechanism and its local load paths. The key engineering questions are whether the shell and support frame remain stiff, whether the raised mechanism can carry retained or jammed regolith, whether connector loads remain modest, and whether asymmetric support creates a critical torsional response.

3 Final CAD Geometry Analyzed

The horizontal configuration contains the final bed shell, its reinforced underside support frame, the rear hinge supports, lower chassis structure, and the active actuator/hinge components included

by each study configuration. The raised configuration rotates the bed by approximately 45° and includes both linear actuators, upper actuator brackets, lower chassis mounts, rear hinge lugs, and six distributed pin connectors.

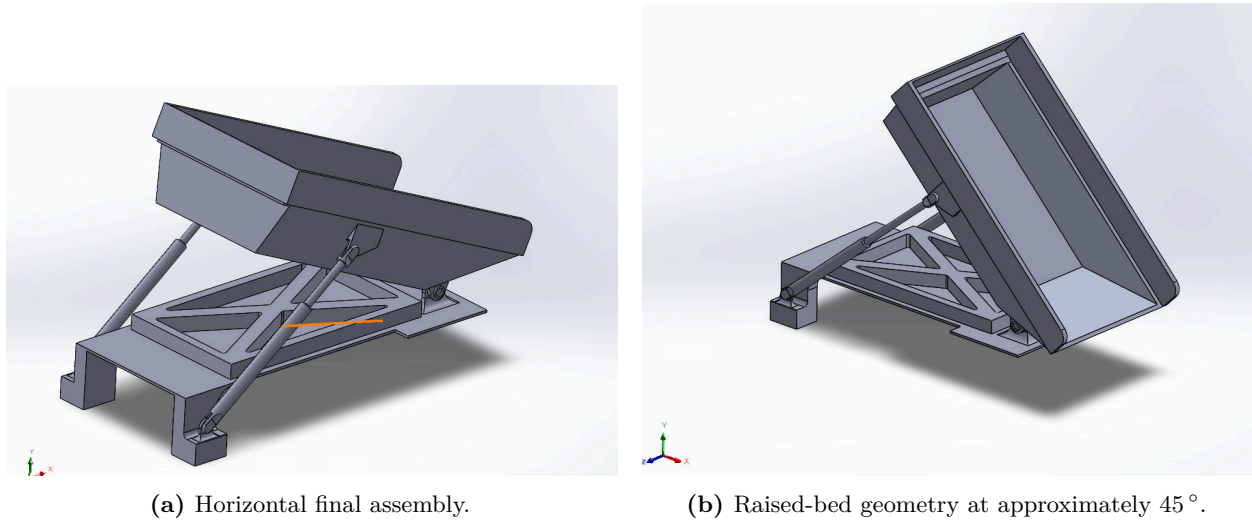


Figure 1: Actual rover dump-bed and chassis CAD configurations used to construct the Week 11 studies.

Table 1 summarizes the principal locked design inputs. The loaded floor area was measured directly from the final CAD rather than using the earlier rectangular estimate.

Table 1: Locked Week 11 geometry and mission inputs.

Parameter	Value	Units
Maximum regolith payload	340	kg
Lunar gravitational acceleration	1.62	m/s ²
Earlier internal bed dimensions	$1.20 \times 0.75 \times 0.35$	m
Earlier geometric bed volume	0.315	m ³
Final CAD loaded floor/lower-surface area	470,691.95	mm ²
Final bed support state	horizontal or $\sim 45^\circ$ raised	–
Nominal parent material	6061-T6 aluminum	–

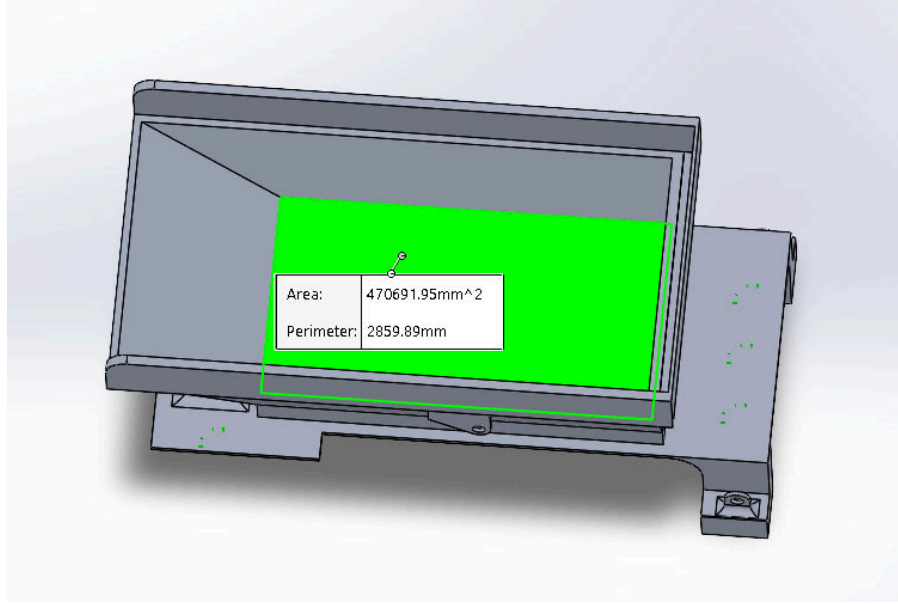


Figure 2: Measured final CAD payload area, 470691.95 mm². This area was used for the pressure calculation.

4 Locked Inputs and Load Calculations

The lunar weight of the full payload is

$$F_p = m_p g_{\text{Moon}} = (340 \text{ kg})(1.62 \text{ m/s}^2) = 550.8 \text{ N}. \quad (1)$$

The measured loaded area is

$$A = 470691.95 \text{ mm}^2 = 0.47069195 \text{ m}^2. \quad (2)$$

Therefore, the exact nominal equivalent pressure is

$$p_1 = \frac{F_p}{A} = \frac{550.8 \text{ N}}{0.47069195 \text{ m}^2} = 1170.19 \text{ Pa} \approx 1170 \text{ N/m}^2. \quad (3)$$

The doubled horizontal design-pressure case is

$$p_2 = 2p_1 \approx 2340 \text{ N/m}^2. \quad (4)$$

For the raised normal case, 25 % retention corresponds to

$$m_{r,25} = 0.25(340 \text{ kg}) = 85 \text{ kg}, \quad (5)$$

$$F_{r,25} = m_{r,25} g_{\text{Moon}} = (85 \text{ kg})(1.62 \text{ m/s}^2) = 137.7 \text{ N}. \quad (6)$$

The abnormal jammed case applies the full retained-payload force of 550.8 N. Gravity at 1.62 m/s² was active in every case, so the modeled structural self-weight was added automatically to the explicit regolith load.

For reference, the complete loaded-rover design load from the Week 11 requirements is

$$W_{\text{rover,design}} = 2(990 \text{ kg})(1.62 \text{ m/s}^2) = 3207.6 \text{ N}. \quad (7)$$

This complete-rover load was not applied to the current dump-bed mechanism studies. It remains a required follow-on full-chassis verification load.

5 Material Properties

All active solid bodies in the final studies were assigned the SOLIDWORKS 6061-T6 (SS) aluminum entry and a linear-elastic isotropic model. Table 2 records the values used. The parent-material yield strength, 275 MPa, is the basis of the automatic von Mises FOS plots.

Table 2: 6061-T6 material properties used in the Week 11 studies.

Property	Value	Units
Elastic modulus, E	6.90×10^{10}	N/m ²
Poisson ratio, ν	0.33	–
Shear modulus, G	2.60×10^{10}	N/m ²
Density, ρ	2700	kg/m ³
Ultimate tensile strength	3.10×10^8	N/m ²
Yield strength, σ_y	2.75×10^8	N/m ²

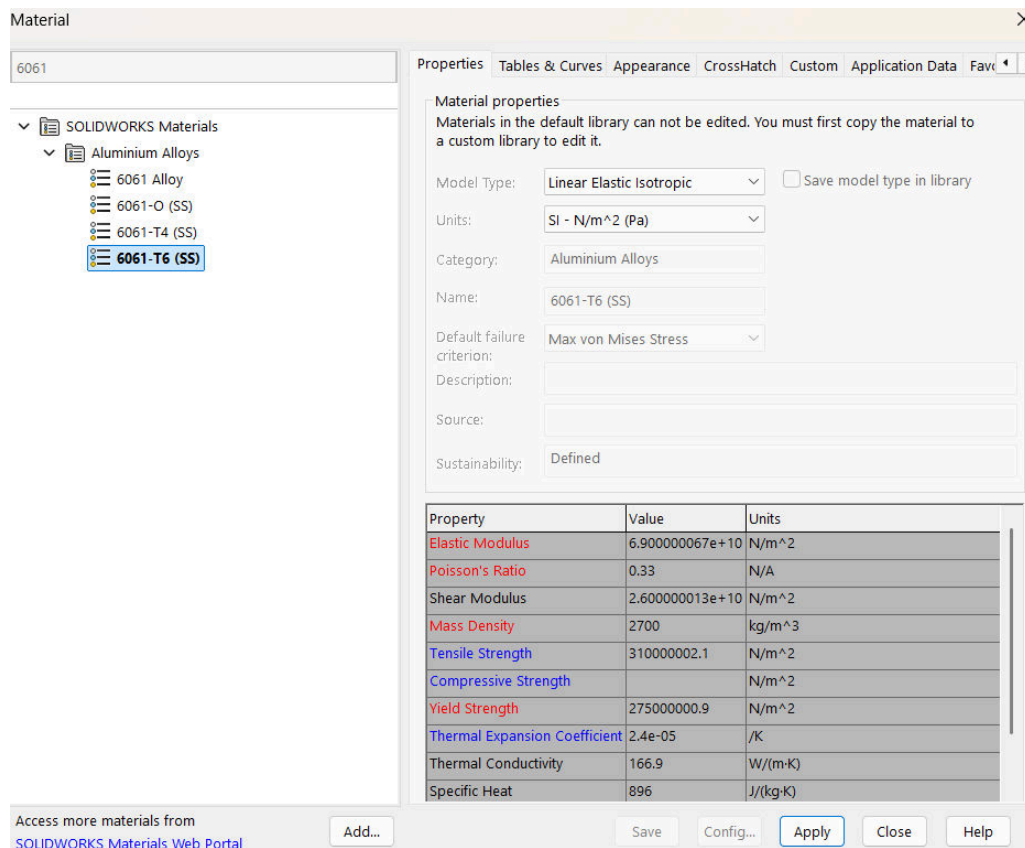


Figure 3: SOLIDWORKS 6061-T6 material record used for the active study bodies.

The model does not reduce strength at welded heat-affected zones, include fastener preload, or model insert and bushing materials. Consequently, the FOS values qualify the idealized 6061-T6 parent material only. Weld allowables, local joint efficiency, fatigue, wear, and manufacturing tolerances require separate verification.

6 Finite-Element Modeling Method

All five cases were solved as linear-static, small-displacement studies. Linear-static analysis assumes slowly applied, time-invariant loads, linear material response, small geometric changes, and boundary conditions that do not change during loading [5,6]. The method is appropriate for preliminary stiffness and yield screening because the reported stresses are orders of magnitude below yield and the displacements are small. It does not model impact, regolith sloshing, contact separation during dumping, actuator dynamics, or plasticity.

The principal modeling choices were as follows:

- **Elements:** high-quality, second-order solid tetrahedral elements with 16-point Jacobian checking.
- **Mesher:** blended curvature-based mesher, which is intended to improve element quality near curvature and difficult assembly geometry [7].
- **Integral/welded structure:** bonded interactions between bodies intended to act as welded or continuous structural load paths.
- **Moving components:** free component interactions between separate bodies, supplemented by local contact where surfaces may bear.
- **Pivots:** distributed pin connectors on coaxial cylindrical faces. Retaining-ring behavior prevented axial translation while rotational stiffness remained zero. Distributed coupling permits local face deformation and reduces artificial rigid-region stress peaks [9,10].
- **Loads:** payload represented as a distributed pressure in horizontal cases and as a vertical distributed force in the raised cases; gravity applied to all active bodies.
- **Results:** von Mises stress, resultant displacement, automatic parent-material FOS, reaction forces, and pin connector forces.

The physical deformation scale shown in the result screenshots is noted on each plot. Several raised-bed plots use a highly exaggerated display scale; the numerical displacement in the legend is the physical value.

7 Boundary-Condition and Connector Strategy

Horizontal nominal and doubled-payload cases

The horizontal assembly was supported at the chassis/base interfaces. Rear hinge rotation was represented by distributed pin connectors. The horizontal bed-to-support interface used a local no-penetration/contact definition where appropriate; bonded interactions represented integral frame and shell connections. Pressure acted normal to the selected internal bed surface, as shown in Figure 4.

Raised-bed cases

The raised configuration included both actuators and six pin connectors: two rear bed hinges, two lower actuator pivots, and two upper actuator pivots. The chassis support faces were fixed for the mechanism screening study. A downward force of 137.7 N (LC3) or 550.8 N (LC4) was applied along

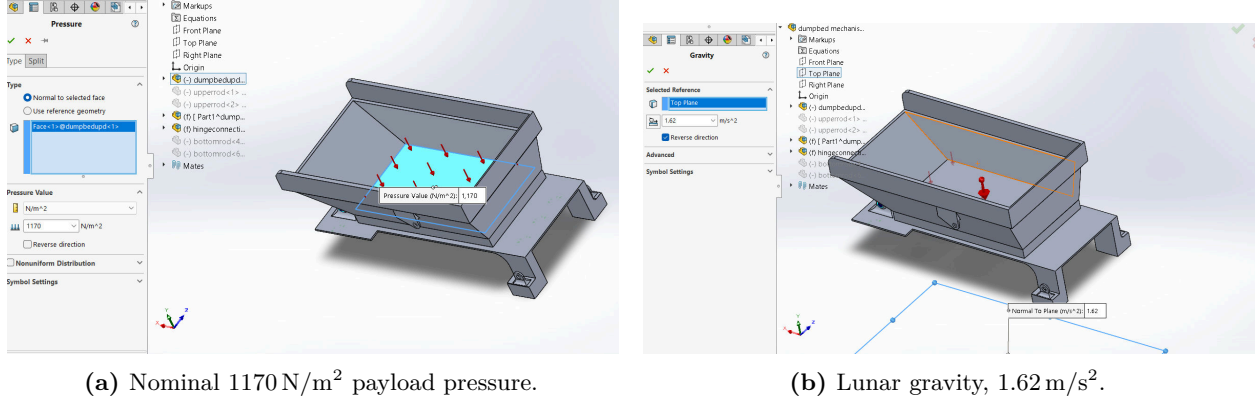


Figure 4: Representative horizontal load definitions.

the global vertical direction to the internal bed floor. This preserves the lunar weight direction rather than incorrectly applying the force normal to the tilted bed.

Asymmetric terrain-support case

LC5 used the horizontal configuration with one fixed support region and two roller/slider support patches. A 100 mm by 100 mm split-face patch localized one terrain reaction. The asymmetry forces the chassis and upper plate to carry torsion instead of receiving symmetric support. The model passed the underconstrained-bodies check before solution.

Fully fixed support faces can locally over-stiffen a real suspension interface and produce edge stress peaks. The roller/slider definitions in LC5 are a better torsional screening idealization, but a later full-rover model should transfer loads through remote constraints, bearing connectors, or suspension-pivot reactions rather than broad fixed faces.

8 Load-Case Definitions

Table 3: Completed Week 11 FEA load cases.

Case	Configuration and purpose	Explicit payload load	Main support idealization
LC1	Horizontal, full payload, nominal lunar condition	1170 N/m ² pressure	Base support plus hinge/contact definitions
LC2	Horizontal, doubled payload-pressure sensitivity	2340 N/m ² pressure	Same as LC1
LC3	Raised $\sim 45^\circ$, normal retained load	137.7 N vertical force	Fixed chassis base, six distributed pins
LC4	Raised $\sim 45^\circ$, abnormal full-payload jam	550.8 N vertical force	Same as LC3
LC5	Horizontal, uneven-terrain torsional support	1170 N/m ² pressure	One fixed and two roller/slider patches

Gravity at 1.62 m/s² was included in every case. LC4 is not a normal operating condition; it is an

intentionally conservative abnormal retained-load case.

9 Mesh Strategy and Quality

Mesh refinement concentrated on hinge bores, actuator-pin holes, bracket fillets, bed-floor transitions, the underside support frame, and local support patches. The horizontal fine study used approximately 6 mm/2.5 mm local controls at the most important faces. Raised studies used 3 mm/1.5 mm controls at pin and lug faces and 7.5 mm/3 mm controls on the bed floor and support frame. LC5 used the same two-control pattern.

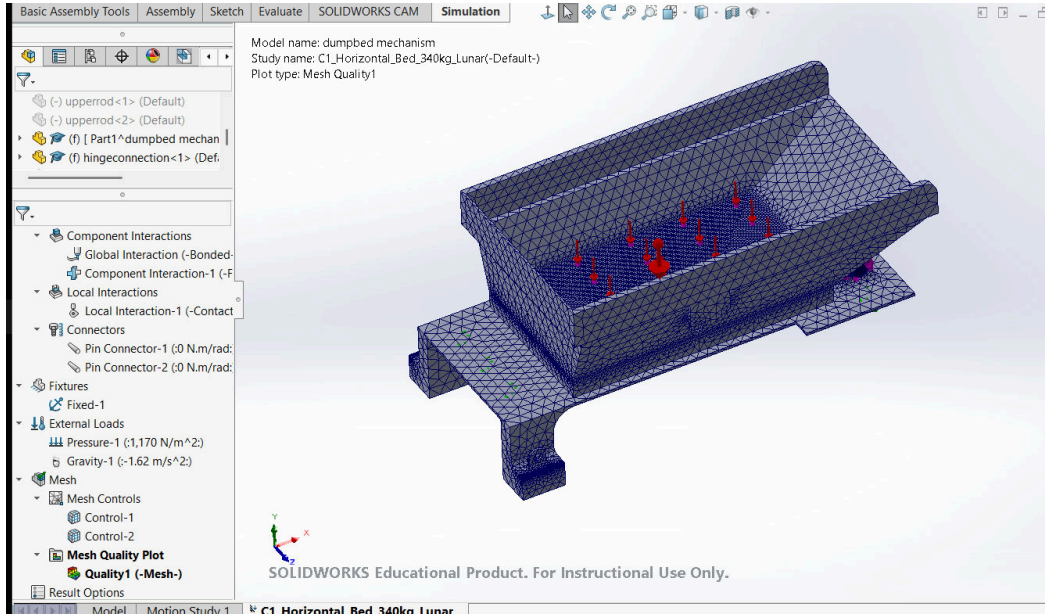


Figure 5: Representative high-quality solid mesh for the integrated horizontal model.

Table 4: Documented global mesh statistics.

Study	Max/min size (mm)	Nodes	Elements	Max AR	AR < 3 (%)	AR > 10 (%)	Distorted
LC1 coarse	40/4	317,325	196,652	201.13	97.1	0.1100	0
LC1 medium	30/3	622,250	400,129	174.54	99.3	0.0517	0
LC1 fine	25/2.5	1,082,744	715,356	140.96	99.6	0.0394	0
LC2 fine	25/2.5	1,082,744	715,356	140.96	99.6	0.0394	0
LC3 raised	30/3	681,874	434,915	174.54	99.2	0.0455	0
LC5 asymmetric	30/3	636,947	406,664	181.34	99.0	0.0843	0

All documented meshes contained zero distorted elements. The maximum aspect ratio occurs in a small fraction of elements; the very small fraction above 10 and the absence of distorted elements support use of the meshes for preliminary screening. SOLIDWORKS identifies Jacobian and aspect-ratio checks as key tetrahedral quality measures [8]. LC4 reused the raised-model control strategy, but a separate final mesh-summary screenshot was not captured; its element count is therefore not invented here.

10 Mesh-Convergence Study

For two consecutive meshes, the reported relative change is

$$\Delta_{i \rightarrow j} = \left| \frac{R_j - R_i}{R_j} \right| (100\%). \quad (8)$$

Table 5 lists the global LC1 result envelopes. The global stress maximum rises with refinement because the maximum occurs at a locally constrained/support edge; it is mesh sensitive. Resultant displacement changes by less than approximately 10 % at each step.

Table 5: LC1 global mesh-convergence results.

Mesh	Nodes	$\sigma_{vm,max}$ (MPa)	u_{max} (mm)	Automatic min FOS
Coarse	317,325	0.052390	0.0006665	5,249
Medium	622,250	0.059210	0.0007306	4,645
Fine	1,082,744	0.072929	0.0008055	3,771

Table 6: Successive LC1 global result changes.

Transition	Stress change (%)	Displacement change (%)
Coarse to medium	11.52	8.77
Medium to fine	18.81	9.30
Coarse to fine	28.16	17.26

To separate fixture-edge behavior from the physical bed response, the stress and displacement were also read on the dump-bed body away from the constrained terrain/support edges. These bed-only values are shown in Table 7.

Table 7: LC1 bed-only convergence away from fixture-edge peaks.

Mesh	Bed stress (MPa)	Change (%)	Bed displacement (mm)	Change (%)
Coarse	0.02649	—	0.0006665	—
Medium	0.02717	2.50	0.0006812	2.16
Fine	0.02888	5.92	0.0006862	0.73

The medium-to-fine bed displacement change is below 1 %, and the bed stress changes by approximately 5.9 %. The coarse-to-fine bed stress change is 8.28 %. This is acceptable for preliminary design screening. The larger global-maximum change is explicitly treated as a local boundary-condition sensitivity rather than evidence of structural yielding.

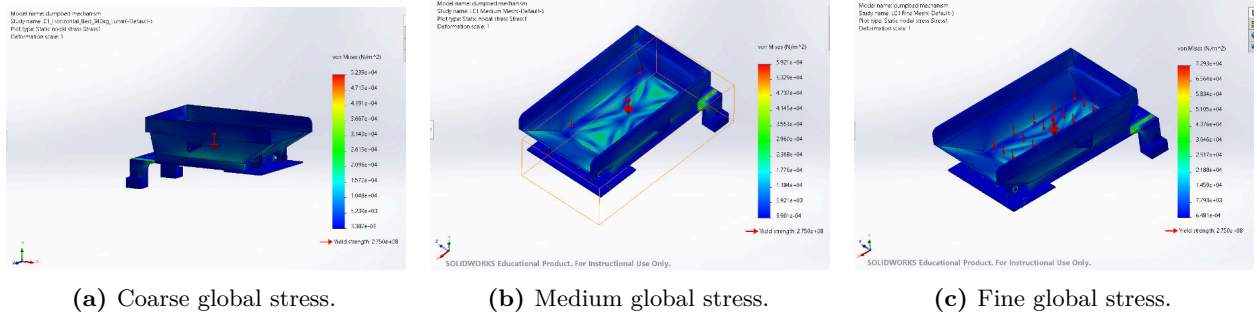


Figure 6: LC1 stress sequence used for the global convergence study.

11 LC1: Horizontal Bed, Nominal Full Payload

LC1 applies the 340 kg lunar payload as 1170 N/m² pressure, together with self-weight under lunar gravity. The fine global maximum von Mises stress was 0.072929 MPa, the maximum resultant displacement was 0.0008055 mm, and the automatic parent-material FOS was 3,771.

For each case,

$$\text{FOS} = \frac{\sigma_y}{\sigma_{vm,\max}}, \quad U = \frac{\sigma_{vm,\max}}{\sigma_y}(100\%). \quad (9)$$

Thus,

$$\text{FOS}_{LC1} = \frac{275 \text{ MPa}}{0.072929 \text{ MPa}} = 3770.8, \quad (10)$$

$$U_{LC1} = \frac{0.072929}{275}(100\%) = 0.0265\%. \quad (11)$$

The bed-only stress away from the fixture edge was 0.02888 MPa, and the corresponding bed-only displacement was 0.0006862 mm. The response is physically reasonable: the floor and long side rails share the distributed load, while the strongest gradients occur near support transitions.

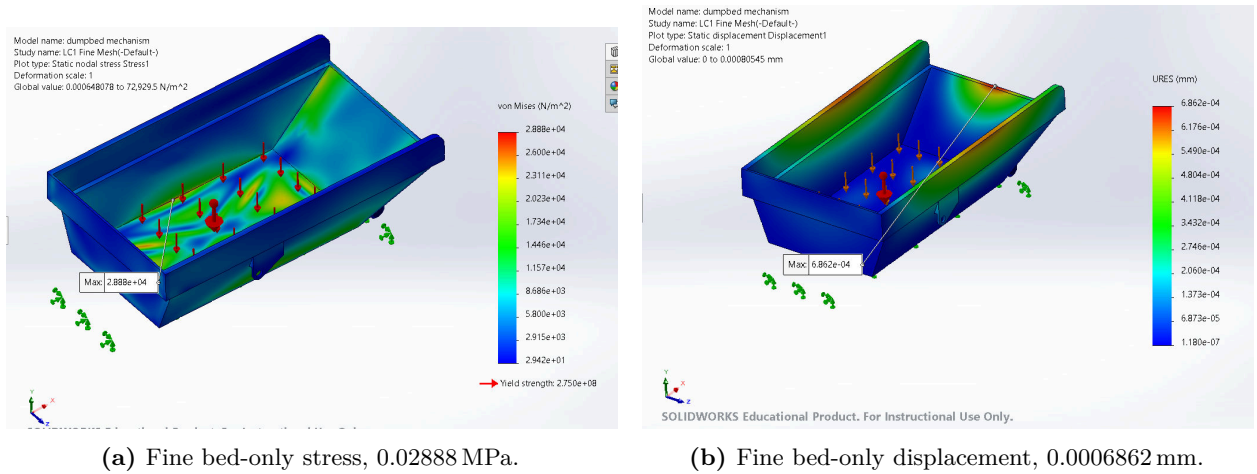


Figure 7: LC1 physical bed response after excluding local fixture-edge peaks.

The reported vertical reaction was 875.3 N. LC1 therefore **PASSES** the parent-material yield criterion with an extremely large margin and negligible displacement.

12 LC2: Horizontal Bed, Doubled Payload Pressure

LC2 retains the horizontal geometry and lunar gravity but doubles the pressure to 2340 N/m². The global maximum von Mises stress was 0.1002 MPa, the reported maximum displacement was 0.0008055 mm, the automatic minimum FOS was 2,745, and the resultant reaction was 1426 N.

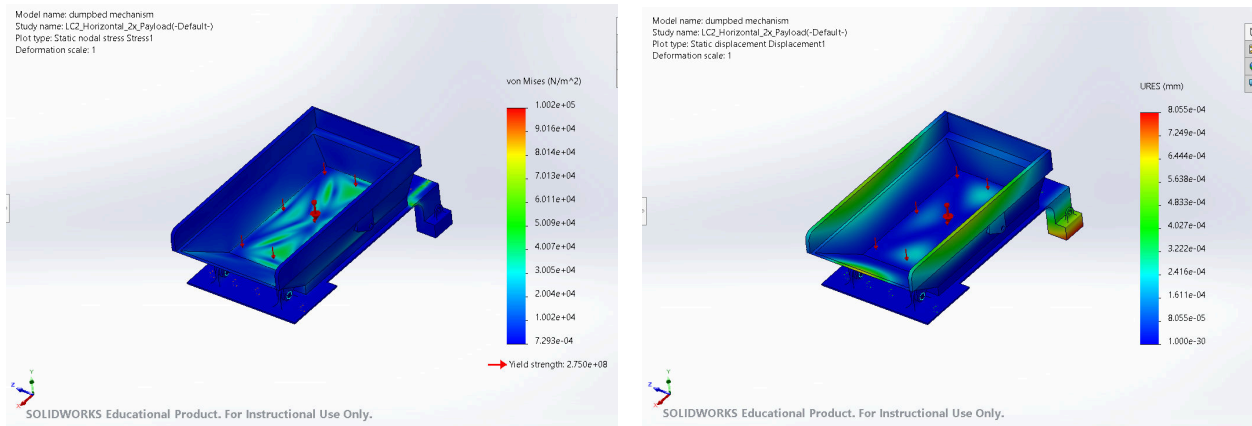
$$FOS_{LC2} = \frac{275}{0.1002} = 2744.5, \quad (12)$$

$$U_{LC2} = \frac{0.1002}{275}(100\%) = 0.0364\%. \quad (13)$$

The added reaction from LC1 to LC2 is

$$\Delta R_{1 \rightarrow 2} = 1426 \text{ N} - 875.3 \text{ N} = 550.7 \text{ N}, \quad (14)$$

which matches the additional 550.8 N payload force caused by doubling the pressure. This is a strong equilibrium check.



(a) Global stress, 0.1002 MPa.

(b) Resultant displacement, 0.0008055 mm.

Figure 8: LC2 doubled-payload results.

The identical displayed global displacement limit in LC1 and LC2 is not used as a linearity proof. It may reflect a constrained-dominated global maximum or a copied result-plot envelope. The reaction change and stress increase confirm that the doubled load was applied. Before final qualification, the displacement plot definition should be regenerated and checked for the active result set. LC2 nevertheless **PASSES** the parent-material yield criterion by more than three orders of magnitude.

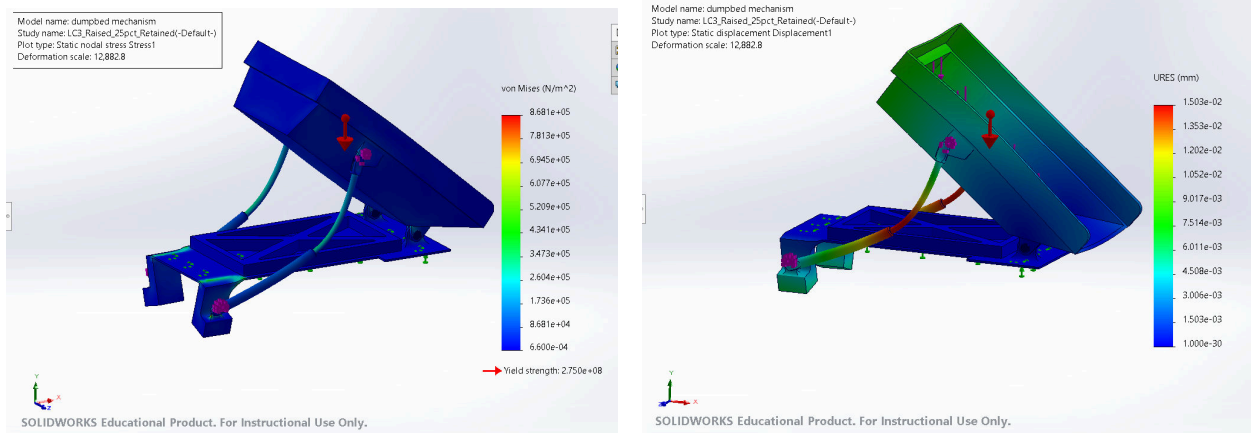
13 LC3: Raised Bed with 25 Percent Retained Payload

LC3 analyzes the bed at approximately 45° with 85 kg of retained regolith, represented by a 137.7 N downward force plus structural self-weight. The maximum von Mises stress was 0.8681 MPa, maximum resultant displacement was 0.01503 mm, and automatic minimum FOS was approximately 320. The resultant support reaction was 473.06 N.

$$FOS_{LC3} = \frac{275}{0.8681} = 316.8, \quad (15)$$

$$U_{LC3} = \frac{0.8681}{275}(100\%) = 0.3157\%. \quad (16)$$

The stress field shows modest local concentrations at the actuator and base attachment regions. The displacement shape is dominated by actuator and bed rotation compliance, with the maximum near the upper end of the raised bed. The plotted deformation scale of 12,882.8 is visual exaggeration; the actual maximum is only 0.01503 mm.



(a) Stress, 0.8681 MPa.

(b) Displacement, 0.01503 mm.

Figure 9: LC3 raised-bed response with 25 percent retained payload.

The governing LC3 connector shear was 84.75 N at Pin Connector 6. The largest reported bending moment was approximately 1.612 N.m at Pin Connector 1. These loads are small relative to typical mechanical hinge pins, but a final pin and lug design still requires actual pin diameter, lug thickness, edge distance, bushing details, and fatigue allowables.

14 LC4: Raised Bed, Abnormal Full-Payload Jam

LC4 is the abnormal conservative case in which the complete 340 kg payload remains in the raised bed. The 550.8 N vertical load acts with structural self-weight. The maximum von Mises stress was 2.137 MPa, maximum displacement was 0.02220 mm, automatic minimum FOS was approximately 130, and resultant reaction was 886.16 N.

$$FOS_{LC4} = \frac{275}{2.137} = 128.7, \quad (17)$$

$$U_{LC4} = \frac{2.137}{275}(100\%) = 0.7771\%. \quad (18)$$

The LC3-to-LC4 reaction increase is

$$\Delta R_{3 \rightarrow 4} = 886.16 \text{ N} - 473.06 \text{ N} = 413.10 \text{ N}, \quad (19)$$

which exactly matches the increase from 25 % to 100 % retained payload:

$$(1.00 - 0.25)(550.8 \text{ N}) = 413.10 \text{ N}. \quad (20)$$

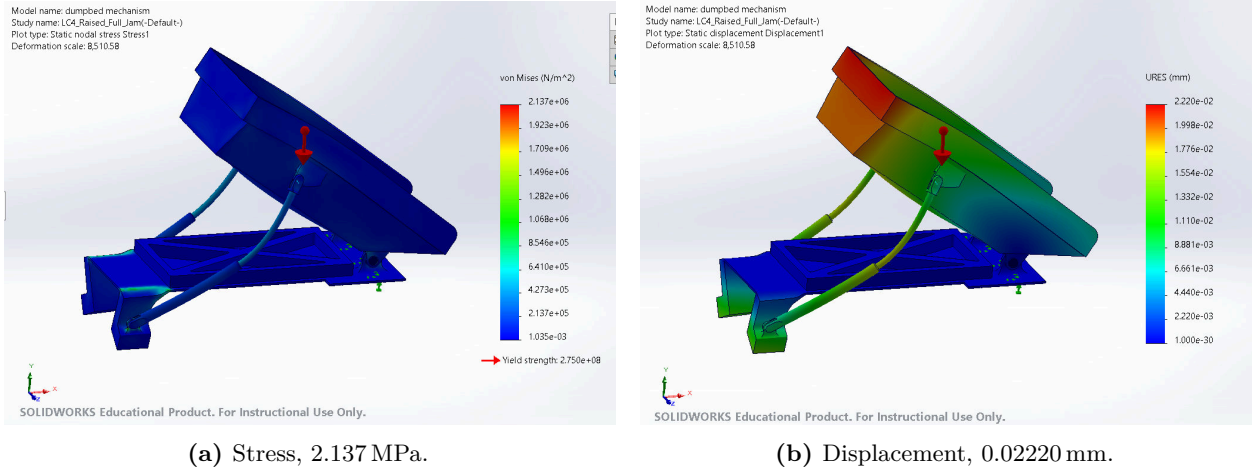


Figure 10: LC4 abnormal full-payload raised-bed response.

The governing connector shear was 208.84 N at Pin Connector 6. The governing connector bending moment was 3.628 N.m at Pin Connector 1, and the largest axial magnitude was approximately 20.63 N at Pin Connector 2. LC4 is the governing raised-bed connector-load case and **PASSes** the parent-material yield criterion.

15 Local Hinge, Actuator-Mount, and Pin Checks

Table 8 summarizes the governing connector results extracted from the solver tables. Numerical torques were on the order of 10^{-13} to 10^{-14} N.m, effectively zero.

Table 8: Governing distributed pin-connector results.

Case	Governing connector quantity	Shear (N)	Axial (N)	Bending (N m)
LC3	Pin 6 shear; Pin 1 bending	84.75	7.20	1.612
LC4	Pin 6 shear; Pin 2 axial; Pin 1 bending	208.84	20.63	3.628
LC5	Pin 1 shear/bending; Pin 2 axial	152.88	14.107	3.623

A conservative hand screen can express the required pin diameter without inventing the final pin size. Using von Mises shear yield and a preliminary safety factor of two,

$$\tau_{\text{allow}} = \frac{0.577\sigma_y}{2} = \frac{0.577(275 \text{ MPa})}{2} = 79.34 \text{ MPa}. \quad (21)$$

For the governing 208.84 N shear and a conservative single-shear plane,

$$d_{\text{req}} = \sqrt{\frac{4V}{\pi\tau_{\text{allow}}}} = \sqrt{\frac{4(208.84 \text{ N})}{\pi(79.34 \text{ MPa})}} = 1.83 \text{ mm}. \quad (22)$$

This very small elastic-shear requirement is only a screening value. Final pins must also satisfy bearing, bending, fatigue, wear, minimum practical diameter, bushing pressure, edge distance, and manufacturing requirements.

For lug bearing with an illustrative parent-material allowable of $\sigma_y/2$,

$$(td)_{\text{req}} = \frac{F}{\sigma_y/2} = \frac{208.84 \text{ N}}{137.5 \text{ MPa}} = 1.52 \text{ mm}^2. \quad (23)$$

Actual lug thickness and hole diameter should be inserted into $\sigma_b = F/(td)$ in the detailed joint drawing. Net-section stress should be checked with $\sigma_{\text{net}} = F/[t(w - d)]$, and tear-out should be checked using the actual edge distance. Weld-group and bracket-bending checks also require the final weld size and load eccentricity.

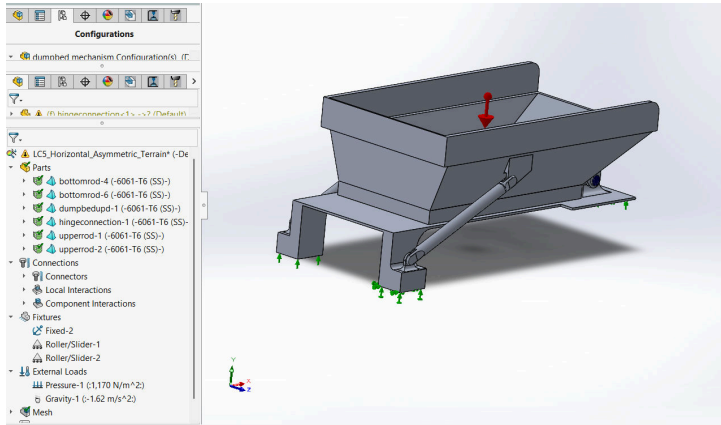
For each actuator at angle θ to the horizontal, a simple symmetric vertical equilibrium lower bound is

$$2F_a \sin \theta + R_{h,y} = W, \quad (24)$$

where $R_{h,y}$ is the vertical hinge reaction. The finite-element connector results are preferred because the hinges and both actuators share load through the actual geometry. A separate actuator buckling and supplier-rating check is still required using the actual extended length, end conditions, and rated force.

16 LC5: Horizontal Asymmetric Terrain Support

LC5 retained the nominal 1170 N/m^2 payload pressure and lunar gravity but changed the ground support to an asymmetric fixed-plus-roller/slider arrangement. The model was fully constrained without fixing the entire underside. The final mesh contained 636,947 nodes and 406,664 elements with zero distorted elements.



(a) Horizontal asymmetric support configuration.

Mesh Details	
Study name	LC5_Horizontal* (LC5_Horizontal-)
DetailsMesh type	Solid Mesh
Mesher Used	Blended curvature-based mesh
Jacobian points for High quality mesh	16 points
Mesh Control	Defined
Max Element Size	30 mm
Min Element Size	3 mm
Mesh quality	High
Total nodes	636947
Total elements	406664
Maximum Aspect Ratio	181.34
Percentage of elements with Aspect Ratio < 3	99
Percentage of elements with Aspect Ratio > 10	0.0843
Percentage of distorted elements	0
Number of distorted elements	0
Remesh failed parts independently	Off
Reuse mesh for identical bodies	Off
Number of bodies that have reused mesh	0
Time to complete mesh(hh:mm:ss)	00:00:11
Computer name	OMKARSPC

(b) Final mesh details.

Figure 11: LC5 geometry, support idealization, and mesh.

The maximum von Mises stress was 5.338 MPa, maximum displacement was 0.1415 mm, and automatic minimum FOS was 52. The resultant reaction was 886.08 N. The peak stress occurs at the edge of the localized terrain-support patch. This is a plausible torsional hotspot but is also

influenced by the abrupt fixture transition; its exact peak magnitude should not be treated as a weld-toe or contact-bearing stress.

$$\text{FOS}_{LC5} = \frac{275}{5.338} = 51.52, \quad (25)$$

$$U_{LC5} = \frac{5.338}{275} (100\%) = 1.941\%. \quad (26)$$

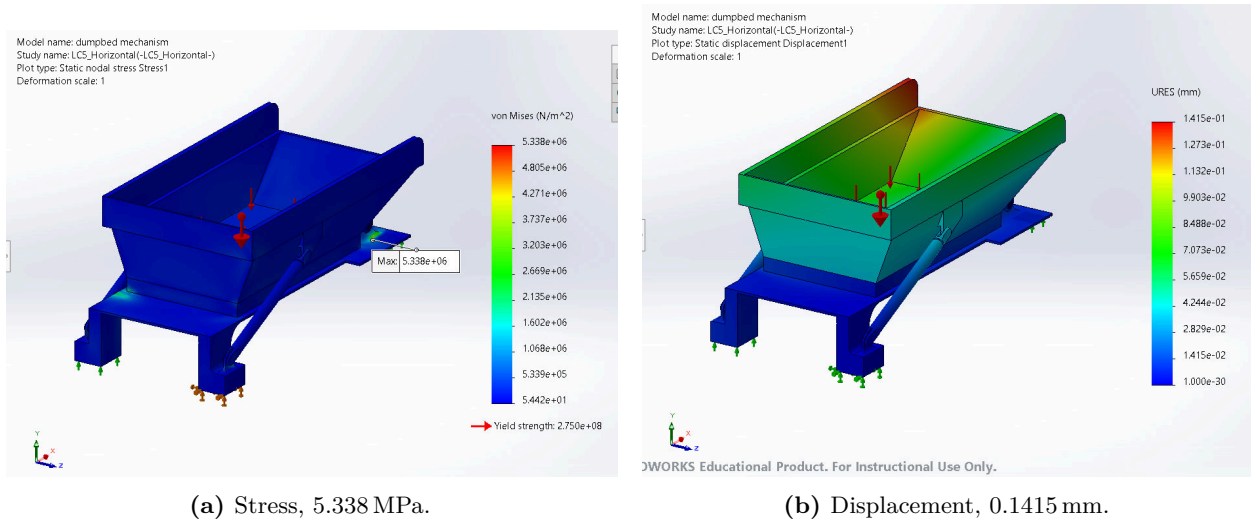


Figure 12: LC5 asymmetric-terrain stress and displacement.

LC5 produces the largest stress and displacement of all completed cases. It is therefore the governing structural case in this report. Even so, the parent-material utilization is below 2%, and the displacement is below 0.15 mm. LC5 **PASS**es the parent-material yield and preliminary stiffness criteria.

17 Overall Results and Structural Pass Assessment

Table 9 compares all completed cases using the global solver maxima. The calculated FOS values use Equation 9; they closely match the automatic SOLIDWORKS plots after rounding.

Table 9: Week 11 global FEA result summary.

Case	Condition	Stress (MPa)	Disp. (mm)	FOS	Util. (%)	Reaction (N)	Status
LC1	Horizontal, nominal payload	0.072929	0.0008055	3770.8	0.0265	875.3	PASS
LC2	Horizontal, doubled pressure	0.1002	0.0008055	2744.5	0.0364	1426	PASS
LC3	Raised, 25% retained	0.8681	0.01503	316.8	0.3157	473.06	PASS
LC4	Raised, full jam	2.137	0.02220	128.7	0.7771	886.16	PASS
LC5	Horizontal, asymmetric terrain	5.338	0.1415	51.52	1.941	886.08	PASS

The governing case is LC5. Its 5.338 MPa maximum stress is only 1.94% of the 275 MPa parent-material yield strength, and its 0.1415 mm displacement is small compared with the assembly dimensions. The governing raised-bed case is LC4, which has a parent-material FOS of approximately

129 and produces the highest pin shear, 208.84 N. All five completed cases pass the preliminary elastic-strength and stiffness checks.

The reaction data provide a useful independent audit. LC1-to-LC2 reactions differ by 550.7 N, exactly the added pressure load. LC3-to-LC4 reactions differ by 413.10 N, exactly the additional 75 % payload. LC4 and LC5 have nearly identical total reactions because both include the full nominal payload force plus similar active structural self-weight, even though their support arrangements and stress fields differ.

An early CAD mass-properties snapshot showed 74.2083 kg with hidden bodies excluded. That value is retained only as configuration traceability; it is not used to back-calculate final study reactions. From reaction balance, the active horizontal LC1/LC2 structure contributes approximately 200.4 kg of modeled gravity mass, while the raised/complete configurations contribute approximately 207.0 kg. The difference is consistent with additional active actuator and linkage bodies. Final documentation should store a mass-properties snapshot for each exact solved configuration.

18 Comparison with Week 7 Hand Calculations

Week 7 recommended a 5 mm 6061-T6 bed floor with transverse support spacing no greater than approximately 0.30 m [2]. The Week 11 CAD contains the final shell, sloped walls, perimeter rails, and reinforced underside support frame, so it is substantially stiffer than the simplified beam-strip idealization.

The horizontal bed-only fine-mesh displacement was approximately 0.000686 mm in LC1. Even the governing asymmetric-terrain displacement was only 0.1415 mm. These values support the Week 7 conclusion that the selected floor thickness and close support spacing are adequate under the analyzed lunar payload. The comparison is qualitative because the Week 7 strip model and the Week 11 three-dimensional assembly have different boundary conditions and load paths.

The Week 7 Earth-handling case was not repeated. No conclusion is made here about a 9.81 m/s^2 ground-handling load, terrestrial lifting, launch vibration, landing shock, or dynamic dumping impact.

19 Design Recommendations

1. **Retain the present shell and support-frame concept.** All completed cases have very low parent-material utilization and small displacement.
2. **Preserve direct load paths at the hinges and actuator mounts.** Brackets should connect into the main rails or reinforced cross-members, not only into broad unsupported sheet.
3. **Add local doublers or gussets where practical.** Although the current stresses are low, these features improve weld load distribution, bearing area, fatigue resistance, and tolerance to local manufacturing defects.
4. **Finalize pin and lug dimensions analytically.** Use the actual pin diameter, lug thickness, edge distance, bushing material, and joint clearance in bearing, net-section, tear-out, bending, and fatigue checks.

5. **Qualify the actuators separately.** Check static force rating, extended-length Euler buckling, clevis capacity, rod-end misalignment, and lunar thermal/lubrication suitability.
6. **Run the complete-rover chassis load case.** Apply the 3207.6 N full-rover design load through representative suspension-pivot reactions and component mass locations before claiming the full chassis is qualified.
7. **Refine the LC5 support interface.** Replace the abrupt support-patch edge with a remote or distributed suspension reaction to test whether the 5.338 MPa hotspot stabilizes.
8. **Recheck result definitions after copied studies.** Regenerate displacement plots and archive mass properties for each active configuration to avoid stale result envelopes.

20 Limitations

The following limitations define the validity of the conclusions:

- Linear-static analysis neglects impact, actuator acceleration, regolith shifting, sloshing, resonance, damping, and contact changes.
- Bonded interactions idealize welded or integral connections and do not calculate weld-throat stress, heat-affected-zone strength, fastener slip, preload, or fatigue.
- Pin connectors idealize the joint and can create local stress concentrations near cylindrical faces. Detailed pin, bushing, and lug contact was not modeled.
- Broad fixed supports and sharp split-face support patches can create artificial stiffness and stress singularities.
- The LC1 global stress maximum is mesh sensitive. The bed-only physical region is substantially better converged.
- The LC2 displayed displacement maximum should be regenerated because it matches LC1 despite the doubled pressure.
- The complete 990 kg rover design load of 3207.6 N was not applied to the full integrated chassis and suspension interfaces.
- No Earth-handling, launch, landing, thermal, modal, fatigue, fracture, regolith abrasion, or lunar-vacuum joint study was completed.
- Automatic FOS plots use the parent 6061-T6 yield strength and do not qualify welds or other materials.

21 Conclusion

The completed Week 11 finite-element studies show that the final dump-bed mechanism and supporting chassis subassembly are structurally adequate for the five analyzed lunar load cases. The nominal horizontal, doubled-pressure, raised 25 %-retention, abnormal raised full-jam, and asymmetric-terrain models all remain in the linear-elastic range with large parent-material margins.

LC5 is the governing case, with 5.338 MPa maximum von Mises stress, 0.1415 mm maximum resultant displacement, and a calculated parent-material FOS of 51.52. LC4 governs the raised-bed connector loads, with 208.84 N maximum pin shear and 3.628 N.m maximum connector bending moment. All

reported displacements are small relative to the mechanism size, and the solver reactions scale correctly with the applied payload changes.

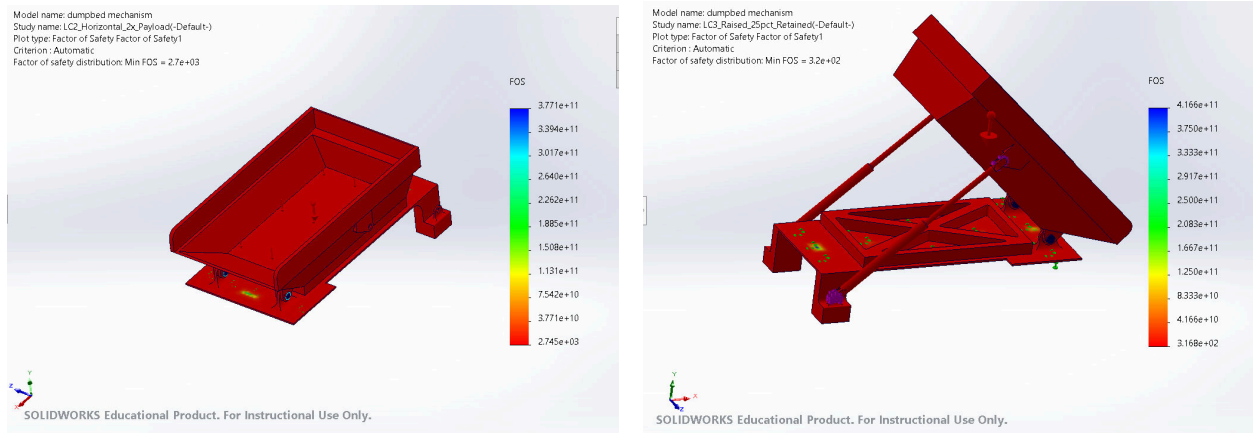
Therefore, the structure **passes the preliminary Week 11 elastic strength and stiffness assessment for the completed load cases**. This pass statement does not constitute final flight qualification. The next structural milestone should be a complete-rover chassis study using the 3207.6 N loaded-rover design load, suspension-pivot reaction inputs, refined weld and joint allowables, and detailed actuator/pin verification.

22 Sources and References

References

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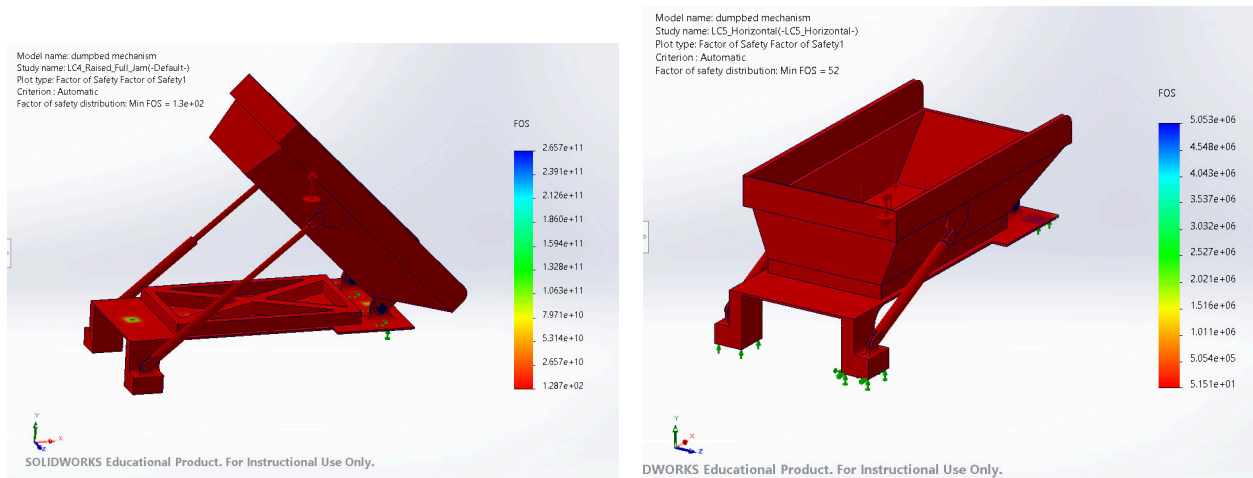
A Supplementary Result Figures



(a) LC2 automatic FOS plot.

(b) LC3 automatic FOS plot.

Figure 13: Automatic parent-material FOS results for LC2 and LC3.



(a) LC4 automatic FOS plot.

(b) LC5 automatic FOS plot.

Figure 14: Automatic parent-material FOS results for LC4 and LC5.

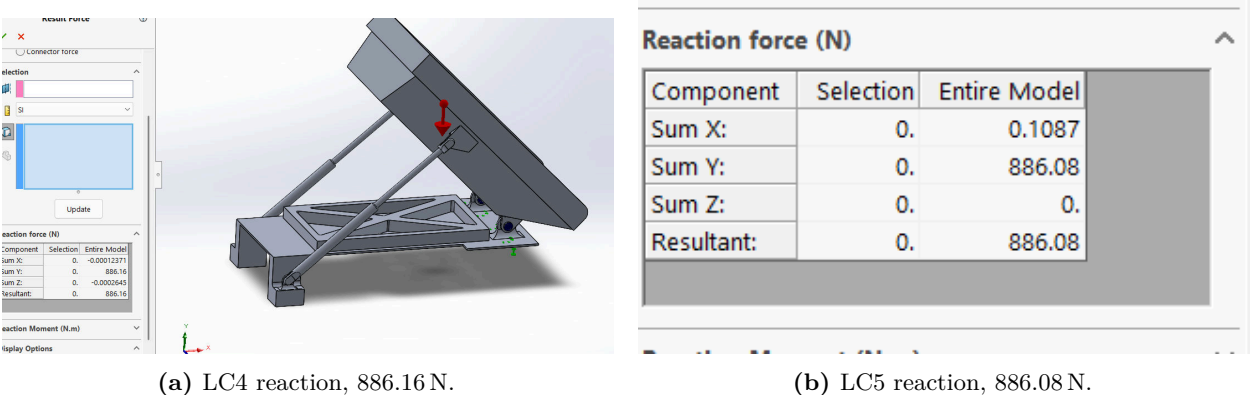


Figure 15: Representative reaction-force result windows.

B Raised-Case Connector Evidence

Type	Resultant	Connector
Shear Force (N)	65.505	Pin Connector-1 Joint 1
Axial Force (N)	5.2305	Pin Connector-1 Joint 1
Bending moment (N.m)	0.34238	Pin Connector-1 Joint 1
Torque (N.m)	-3.399e-14	Pin Connector-1 Joint 1
Shear Force (N)	65.505	Pin Connector-1 Joint 2
Axial Force (N)	5.2305	Pin Connector-1 Joint 2
Bending moment (N.m)	1.6116	Pin Connector-1 Joint 2
Torque (N.m)	1.2787e-14	Pin Connector-1 Joint 2
Shear Force (N)	65.56	Pin Connector-1 Joint 3
Axial Force (N)	-7.1394	Pin Connector-1 Joint 3
Bending moment (N.m)	0.35585	Pin Connector-1 Joint 3
Torque (N.m)	-3.502e-14	Pin Connector-1 Joint 3

Figure 16: LC3 connector-force listing for Pin Connector 1.

Type	Resultant	Connector
Torque (N.m)	4.9641e-13	Pin Connector-3 Joint 1
Shear Force (N)	81.168	Pin Connector-3 Joint 2
Axial Force (N)	-0.055359	Pin Connector-3 Joint 2
Bending moment (N.m)	0.05659	Pin Connector-3 Joint 2
Torque (N.m)	6.227e-14	Pin Connector-3 Joint 2
Shear Force (N)	81.328	Pin Connector-4 Joint 1
Axial Force (N)	0.077166	Pin Connector-4 Joint 1
Bending moment (N.m)	0.054566	Pin Connector-4 Joint 1
Torque (N.m)	6.1433e-14	Pin Connector-4 Joint 1
Shear Force (N)	81.328	Pin Connector-4 Joint 2
Axial Force (N)	-0.077166	Pin Connector-4 Joint 2
Bending moment (N.m)	0.054773	Pin Connector-4 Joint 2
Torque (N.m)	4.9579e-13	Pin Connector-4 Joint 2

Type	Resultant	Connector
Torque (N.m)	2.6467e-13	Pin Connector-5 Joint 1
Shear Force (N)	84.592	Pin Connector-5 Joint 2
Axial Force (N)	-0.055372	Pin Connector-5 Joint 2
Bending moment (N.m)	0.11493	Pin Connector-5 Joint 2
Torque (N.m)	-9.64e-14	Pin Connector-5 Joint 2
Shear Force (N)	84.752	Pin Connector-6 Joint 1
Axial Force (N)	-0.077164	Pin Connector-6 Joint 1
Bending moment (N.m)	0.10257	Pin Connector-6 Joint 1
Torque (N.m)	9.6253e-14	Pin Connector-6 Joint 1
Shear Force (N)	84.752	Pin Connector-6 Joint 2
Axial Force (N)	0.077164	Pin Connector-6 Joint 2
Bending moment (N.m)	0.13558	Pin Connector-6 Joint 2
Torque (N.m)	-2.6572e-13	Pin Connector-6 Joint 2

(a) LC3 middle connectors.

(b) LC3 Pin Connectors 5 and 6.

Figure 17: LC3 connector-force evidence.

Connector Force					
Type	Resultant	Connector	Type	Resultant	Connector
Shear Force (N)	147.3	Pin Connector-1 Joint 1	Torque (N.m)	-9.1845e-14	Pin Connector-2 Joint 1
Axial Force (N)	8.9463	Pin Connector-1 Joint 1	Shear Force (N)	143.1	Pin Connector-2 Joint 2
Bending moment (N.m)	0.77197	Pin Connector-1 Joint 1	Axial Force (N)	20.63	Pin Connector-2 Joint 2
Torque (N.m)	-9.2145e-14	Pin Connector-1 Joint 1	Bending moment (N.m)	3.5187	Pin Connector-2 Joint 2
Shear Force (N)	147.3	Pin Connector-1 Joint 2	Torque (N.m)	3.2046e-14	Pin Connector-2 Joint 2
Axial Force (N)	8.9463	Pin Connector-1 Joint 2	Shear Force (N)	147.69	Pin Connector-2 Joint 3
Bending moment (N.m)	3.6281	Pin Connector-1 Joint 2	Axial Force (N)	-9.0276	Pin Connector-2 Joint 3
Torque (N.m)	3.1976e-14	Pin Connector-1 Joint 2	Bending moment (N.m)	0.7726	Pin Connector-2 Joint 3
Shear Force (N)	143.14	Pin Connector-1 Joint 3	Torque (N.m)	-9.0515e-14	Pin Connector-2 Joint 3
Axial Force (N)	-20.496	Pin Connector-1 Joint 3	Shear Force (N)	205.2	Pin Connector-3 Joint 1
Bending moment (N.m)	0.77221	Pin Connector-1 Joint 3	Axial Force (N)	0.091614	Pin Connector-3 Joint 1
Torque (N.m)	-9.3448e-14	Pin Connector-1 Joint 3	Bending moment (N.m)	0.16182	Pin Connector-3 Joint 1
Shear Force (N)	143.1	Pin Connector-2 Joint 1	Torque (N.m)	5.1196e-13	Pin Connector-3 Joint 1
Axial Force (N)	20.63	Pin Connector-2 Joint 1	Shear Force (N)	205.2	Pin Connector-3 Joint 2

(a) LC4 Pin Connector 1.

(b) LC4 Pin Connectors 2 and 3.

Figure 18: LC4 connector-force evidence, part 1.

Connector Force			Connector Force		
Type	Resultant	Connector	Type	Resultant	Connector
Shear Force (N)	205.2	Pin Connector-3 Joint 2	Shear Force (N)	208.62	Pin Connector-5 Joint 1
Axial Force (N)	-0.091614	Pin Connector-3 Joint 2	Axial Force (N)	0.091639	Pin Connector-5 Joint 1
Bending moment (N.m)	0.16119	Pin Connector-3 Joint 2	Bending moment (N.m)	0.25961	Pin Connector-5 Joint 1
Torque (N.m)	1.6054e-13	Pin Connector-3 Joint 2	Torque (N.m)	2.3896e-13	Pin Connector-5 Joint 1
Shear Force (N)	205.42	Pin Connector-4 Joint 1	Shear Force (N)	208.62	Pin Connector-5 Joint 2
Axial Force (N)	0.14481	Pin Connector-4 Joint 1	Axial Force (N)	-0.091639	Pin Connector-5 Joint 2
Bending moment (N.m)	0.15596	Pin Connector-4 Joint 1	Bending moment (N.m)	0.25816	Pin Connector-5 Joint 2
Torque (N.m)	1.5872e-13	Pin Connector-4 Joint 1	Torque (N.m)	-2.4837e-13	Pin Connector-5 Joint 2
Shear Force (N)	205.42	Pin Connector-4 Joint 2	Shear Force (N)	208.84	Pin Connector-6 Joint 1
Axial Force (N)	-0.14481	Pin Connector-4 Joint 2	Axial Force (N)	-0.14483	Pin Connector-6 Joint 1
Bending moment (N.m)	0.15652	Pin Connector-4 Joint 2	Bending moment (N.m)	0.2291	Pin Connector-6 Joint 1
Torque (N.m)	5.1044e-13	Pin Connector-4 Joint 2	Torque (N.m)	2.3839e-13	Pin Connector-6 Joint 1
Shear Force (N)	208.62	Pin Connector-5 Joint 1	Shear Force (N)	208.84	Pin Connector-6 Joint 2
Axial Force (N)	0.091639	Pin Connector-5 Joint 1	Axial Force (N)	0.14483	Pin Connector-6 Joint 2

(a) LC4 Pin Connectors 3 and 4.

(b) LC4 Pin Connectors 5 and 6.

Figure 19: LC4 connector-force evidence, part 2.

Type	Resultant	Connector
Shear Force (N)	152.88	Pin Connector-1 Joint 1
Axial Force (N)	0.19278	Pin Connector-1 Joint 1
Bending moment (N.m)	1.0501	Pin Connector-1 Joint 1
Torque (N.m)	-5.1318e-14	Pin Connector-1 Joint 1
Shear Force (N)	152.88	Pin Connector-1 Joint 2
Axial Force (N)	0.19278	Pin Connector-1 Joint 2
Bending moment (N.m)	3.6232	Pin Connector-1 Joint 2
Torque (N.m)	6.2598e-13	Pin Connector-1 Joint 2
Shear Force (N)	141.59	Pin Connector-1 Joint 3
Axial Force (N)	-9.837	Pin Connector-1 Joint 3
Bending moment (N.m)	0.99714	Pin Connector-1 Joint 3
Torque (N.m)	-6.7675e-14	Pin Connector-1 Joint 3
Shear Force (N)	83.833	Pin Connector-2 Joint 1
Axial Force (N)	14.107	Pin Connector-2 Joint 1

(a) LC5 Pin Connector 1.

Type	Resultant	Connector
Torque (N.m)	-6.7675e-14	Pin Connector-1 Joint 3
Shear Force (N)	83.833	Pin Connector-2 Joint 1
Axial Force (N)	14.107	Pin Connector-2 Joint 1
Bending moment (N.m)	0.37649	Pin Connector-2 Joint 1
Torque (N.m)	7.897e-13	Pin Connector-2 Joint 1
Shear Force (N)	83.833	Pin Connector-2 Joint 2
Axial Force (N)	14.107	Pin Connector-2 Joint 2
Bending moment (N.m)	2.4893	Pin Connector-2 Joint 2
Torque (N.m)	9.3186e-13	Pin Connector-2 Joint 2
Shear Force (N)	38.102	Pin Connector-2 Joint 3
Axial Force (N)	-11.222	Pin Connector-2 Joint 3
Bending moment (N.m)	0.72834	Pin Connector-2 Joint 3
Torque (N.m)	7.6398e-13	Pin Connector-2 Joint 3

(b) LC5 Pin Connector 2.

Figure 20: LC5 horizontal asymmetric-terrain connector-force evidence.

C Traceability Screenshots

Mesh Details	
Study name	LC1 Fine Mesh* (-Default-)
DetailsMesh type	Solid Mesh
Mesher Used	Blended curvature-based mesh
Jacobian points for High quality mesh	16 points
Mesh Control	Defined
Max Element Size	25 mm
Min Element Size	2.5 mm
Mesh quality	High
Total nodes	1082744
Total elements	715356
Maximum Aspect Ratio	140.96
Percentage of elements with Aspect Ratio < 3	99.6
Percentage of elements with Aspect Ratio > 10	0.0394
Percentage of distorted elements	0
Number of distorted elements	0
Remesh failed parts independently	Off
Reuse mesh for identical bodies	Off
Number of bodies that have reused mesh	0
Time to complete mesh(hh:mm:ss)	00:00:13
Computer name	OMKARSPC

Mesh Details	
Study name	LC3_Raised_25pct_Retained* (-Default-)
DetailsMesh type	Solid Mesh
Mesher Used	Blended curvature-based mesh
Jacobian points for High quality mesh	16 points
Mesh Control	Defined
Max Element Size	30 mm
Min Element Size	3 mm
Mesh quality	High
Total nodes	681874
Total elements	434915
Maximum Aspect Ratio	174.54
Percentage of elements with Aspect Ratio < 3	99.2
Percentage of elements with Aspect Ratio > 10	0.0455
Percentage of distorted elements	0
Number of distorted elements	0
Remesh failed parts independently	Off
Reuse mesh for identical bodies	Off
Number of bodies that have reused mesh	0
Time to complete mesh(hh:mm:ss)	00:00:14
Computer name	OMKARSPC

(a) LC1 fine-mesh summary.

(b) LC3 raised-mesh summary.

Figure 21: Mesh-summary evidence used in Tables 4 and 5.

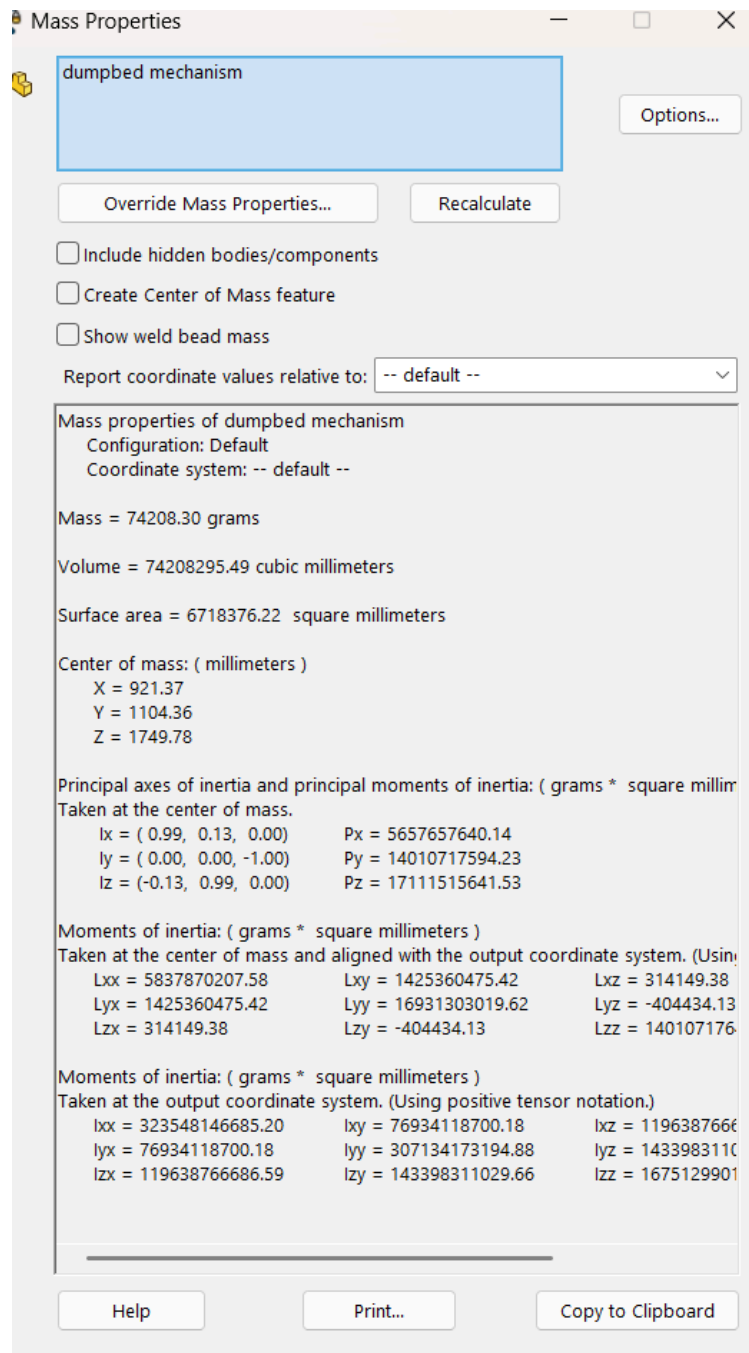


Figure 22: Early CAD mass-properties snapshot. Hidden bodies were excluded; the result is retained for traceability and is not used as the final active-study mass.